

Automated minus reference counts on the gold standard: protocol change

tranche mixed_gold_cont120_02 - frozen baseline versus candidate protocol on the same manifest and gold - misses and abstentions scored as 0 - square-root axes - every asserted gold row

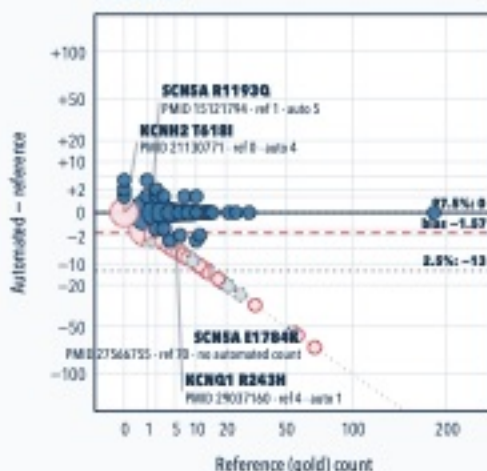
● automated count supplied ○ variant matched, no automated count (scored 0) ○ variant not found (scored 0)

Marker area - gold rows at one coordinate (wedges split it by status):

○ 1 ○ 8 ○ 28

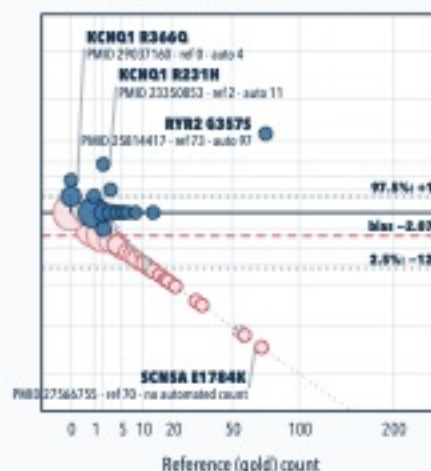
Baseline arm - 20260905_protocol_cont120_02_baseline - runtime e620f992 - locked 2026-09-05 - 120 gene-paper attempts

Carriers



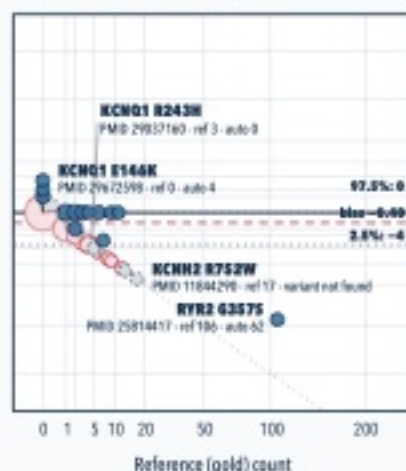
571 gold rows - matched 77.2% - supplied 49.4%
exact 54.5% (zero-baseline 8.6%) - within ± 1 81.6%
bias -1.57 - c2e MAE 1.634 - cond. MAE 0.121 (n 282)
empirical 95% limits -13 to 0 - RMSE 5.64

Affected individuals



571 gold rows - matched 77.2% - supplied 18.7%
exact 31.5% (zero-baseline 16.8%) - within ± 1 67.4%
bias -2.07 - c2e MAE 2.256 - cond. MAE 0.542 (n 187)
empirical 95% limits -12 to +1 - RMSE 5.97

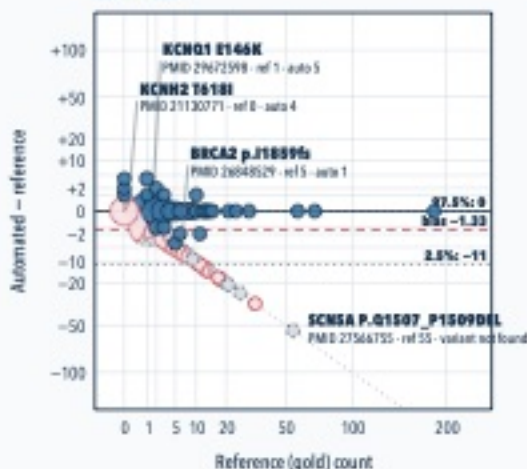
Unaffected individuals



571 gold rows - matched 77.2% - supplied 6.0%
exact 86.7% (zero-baseline 82.3%) - within ± 1 93.5%
bias -0.40 - c2e MAE 0.426 - cond. MAE 1.765 (n 34)
empirical 95% limits -4 to 0 - RMSE 2.33

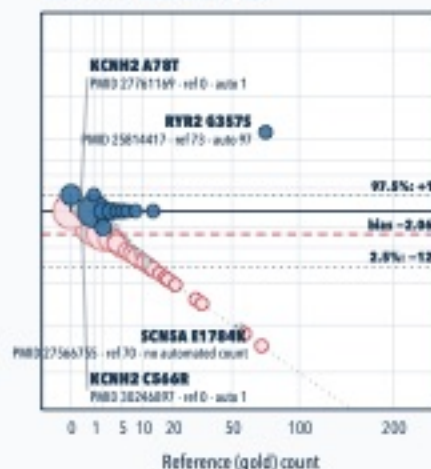
Candidate arm - 20260905_protocol_cont120_02_candidate - runtime e9fef8eb - locked 2026-09-05 - 120 gene-paper attempts

Carriers



571 gold rows - matched 77.9% - supplied 51.3%
exact 56.9% (zero-baseline 8.6%) - within ± 1 81.8%
bias -1.33 - c2e MAE 1.405 - cond. MAE 0.113 (n 293)
empirical 95% limits -11 to 0 - RMSE 4.18

Affected individuals



571 gold rows - matched 77.9% - supplied 21.4%
exact 34.0% (zero-baseline 16.8%) - within ± 1 68.3%
bias -2.06 - c2e MAE 2.281 - cond. MAE 0.377 (n 122)
empirical 95% limits -12 to +1 - RMSE 5.95

Unaffected individuals



571 gold rows - matched 77.9% - supplied 7.2%
exact 87.2% (zero-baseline 82.0%) - within ± 1 93.9%
bias -0.39 - c2e MAE 0.403 - cond. MAE 1.415 (n 41)
empirical 95% limits -4 to 0 - RMSE 2.29

Candidate - baseline on the same 1713 gold rows (paired by gold row; deltas are candidate minus baseline)

Carriers

c2e MAE 1.634 $\rightarrow$ 1.405 (-0.229) - registered UB +0.064
supplied 282 $\rightarrow$ 293 - exact 311 $\rightarrow$ 325 - bias -1.57 $\rightarrow$ -1.33
rows better 27 - worse 16 - unchanged 528
coverage on matched 63.9% $\rightarrow$ 63.8% (+1.90 pp) - LB -1.94 pp

Affected individuals

c2e MAE 2.256 $\rightarrow$ 2.201 (-0.054)
supplied 187 $\rightarrow$ 122 - exact 180 $\rightarrow$ 194 - bias -2.07 $\rightarrow$ -2.06
rows better 32 - worse 16 - unchanged 523
coverage on matched 24.3% $\rightarrow$ 27.4% (+3.15 pp)

Unaffected individuals

c2e MAE 0.426 $\rightarrow$ 0.403 (-0.023)
supplied 34 $\rightarrow$ 41 - exact 495 $\rightarrow$ 498 - bias -0.40 $\rightarrow$ -0.39
rows better 7 - worse 2 - unchanged 562
coverage on matched 7.7% $\rightarrow$ 9.2% (+1.50 pp)

Identity (primary endpoint): recall 77.23% $\rightarrow$ 77.93% (+0.70 pp; LB -0.17 pp) - precision 70.79% $\rightarrow$ 71.20% (+0.41 pp; LB -1.29 pp) - registered decision: reject_or_revise_candidate - secondary counts